THE UPPER SHRIEK OF THE STRUCTURE SHEAF ALONG AN LCI MORPHISM IS A SHIFTED INVERTIBLE SHEAF

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Allowed standard prerequisites (editorial).

- The local factorization of an lci morphism into a smooth morphism followed by a regular closed immersion. Used inside the counted proof at lines 694-695. Reduces the statement to a smooth morphism g and a regular closed immersion iota, for which the upper shriek is computed separately.
- The composition formula for the upper shriek and the identification of g^! of the structure sheaf with shifted m-differentials for a smooth morphism. Used inside the counted proof at lines 696-700. Produces the displayed chain of equalities for f^! of the structure sheaf.
- The Koszul complex of a regular sequence is a flat resolution, and its dual resolves the quotient shifted by the codimension. Used inside the counted proof at lines 701-707. Replaces the derived hom by a tensor computation

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and identifies the result with the restriction of the invertible sheaf of m-differentials.

Lemma 0.1. *Let $f : X \to Y$ be an lci morphism of relative dimension n between locally noetherian schemes of finite Krull dimension. Then $f^!\mathscr{O}_Y = \omega^\circ_{X/Y}[n]$ with $\omega^\circ_{X/Y}$ an invertible $\mathscr{O}_X$-module.*

Proof. Working locally on Y and X, we may assume that f factors as $X \xrightarrow{\iota} S \xrightarrow{g} Y$, where g is a smooth morphism of relative dimension m, and ι is a regular closed immersion of codimension $m-n$ defined by an ideal $\mathscr{I} = (f_1, \dots, f_{m-n})$. Let $\omega^\circ_{S/Y} = \bigwedge^m \Omega^1_{S/Y}$ be the sheaf of m-differentials of S over Y, then

$$\begin{aligned} f^!\mathscr{O}_Y &= \iota^! g^! \mathscr{O}_Y \\ &= \iota^{-1} \mathrm{R}\mathscr{H}om_{\mathscr{O}_S}(\iota_*\mathscr{O}_X, \omega^\circ_{S/Y}[m]) \\ &= \iota^{-1} \mathrm{R}\mathscr{H}om_{\mathscr{O}_S}(\mathscr{O}_S/\mathscr{I}, \omega^\circ_{S/Y})[m]. \end{aligned}$$

Let $K(\underline{f})$ be the Koszul complex of the regular sequence $\underline{f} = (f_1, \dots, f_{m-n})$. Then $K(\underline{f})$ is a flat resolution of $\mathscr{O}_S/\mathscr{I}$, its dual $K(\underline{f})^\vee = \mathscr{H}om_{\mathscr{O}_S}(K(\underline{f}), \mathscr{O}_S)$ is a flat resolution of $(\mathscr{O}_S/\mathscr{I})[-(m-n)]$. Therefore

$$\begin{aligned} f^!\mathscr{O}_Y &\simeq \iota^{-1} \mathscr{H}om_{\mathscr{O}_S}(K(\underline{f}), \omega^\circ_{S/Y})[m] \\ &= \iota^{-1} K(\underline{f})^\vee \otimes \omega^\circ_{S/Y}[m] \\ &\simeq \iota^{-1}(\mathscr{O}_S/\mathscr{I} \otimes^L \omega^\circ_{S/Y}[n]) \\ &= \iota^{-1}((\omega^\circ_{S/Y}/\mathscr{I})[n]) = \iota^*\omega^\circ_{S/Y}[n], \end{aligned}$$

which is an invertible sheaf of $\mathscr{O}_X$-modules as required. □

References

[1] R. Hartshorne - Residues and duality , Lect. Notes in Math. , vol. 20 , Springer-Verlag, Berlin-New York, 1966

[2] D. Mumford - Abelian varieties , TIFR Studies in Mathematics , vol. 5 , Tata Institute of Fundamental Research, Bombay, 2008